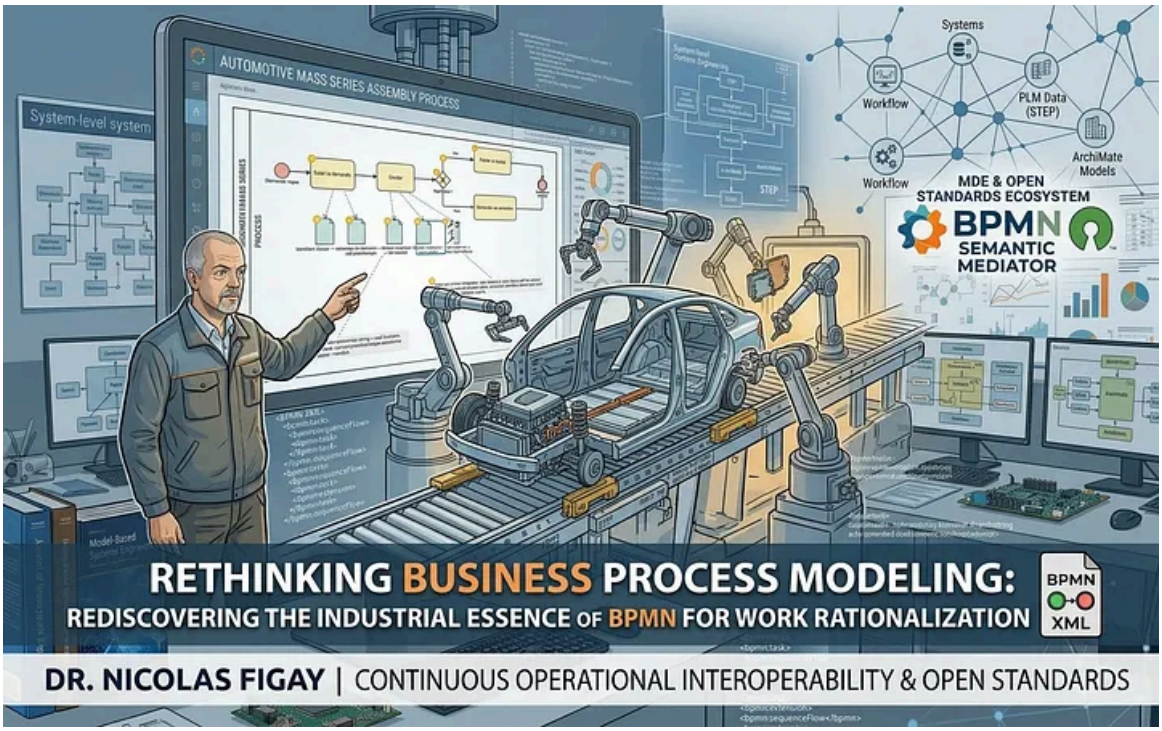


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Rethinking Business Process Modeling: Rediscovering the Industrial Essence of BPMN for Work Rationalization

Dr Nicolas Figay Follow 5 min read · 1 day ago

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In an era dominated by agile frameworks, ad-hoc workflows, and digital service interactions, the foundational core of business process management

is sometimes perceived as a remnant of an earlier industrial age. Modern literature often questions the relevance of formal modeling in **BPMN** (*Business Process Model and Notation*), dismissing explicit process specifications as overly rigid or disconnected from dynamic market demands. However, this skepticism overlooks a fundamental historical truth: the true value of process management lies in its ability to **industrialize operations** — transforming chaotic, craft-based (*artisanat*) activities into systematic, repeatable, and scalable series production. To understand why BPMN remains an indispensable instrument in enterprise architecture, we must re-anchor process modeling in its core purpose: the **rationalization and engineering of work**.

1. What is the Historical Connection Between Process Management and Industrialization?

The formal concept of a process emerged directly from the First and Second Industrial Revolutions to solve a critical economic challenge: transitioning from custom, artisan craftsmanship to mass series production of complex, highly sophisticated goods (such as automotive vehicles or aerospace equipment). Early pioneers of Scientific Management, notably **Frederick W. Taylor** (*The Principles of Scientific Management*, 1911) and **Henry Ford** with the assembly line in 1913, demonstrated that mass production required breaking down complex endeavors into discrete, standardized, and measurable sequence steps. Process definition became the foundational asset that enabled scaling output, reducing variance, and ensuring consistent product quality.

2. How Does Craftsmanship (Artisanat) Differ from Industrialized Process Execution?

Craftsmanship relies primarily on **implicit tacit knowledge**, individual skills, and ad-hoc decision-making localized within a single artisan. While highly flexible, craft-based execution cannot support the mass production of sophisticated products due to non-reproducible quality, extreme dependency on scarce experts, and an inability to scale across large organizations. In contrast, **industrialized execution** decouples operational know-how from individual memory by encoding it into an explicit, shared meta-model. This transition requires formalizing sequence flows, resource

allocations, and business rules so that execution can be predictably repeated and optimized.

3. Why is the Industrial Heritage of Business Process Management Often Forgotten Today?

With the rise of the service economy, knowledge work, and cloud-native software platforms, many organizations began viewing process management purely as an administrative or documentation burden. The shift toward unstructured case management and dynamic ad-hoc tools led some analysts to declare formal process modeling obsolete, arguing that modern work is too unpredictable for rigid flows. However, this view mistakes **process execution for bureaucracy**. Without an explicit process architecture, digital transformation initiatives frequently degenerate into digitizing chaotic, craft-based practices — leading to duplicated efforts, unintegrated software silos, and severe operational fragility.

4. How Does BPMN Act as an Engineering Tool for Work Rationalization?

Far from being a mere drawing standard, **BPMN 2.0** provides a rigorous mathematical and logical grammar designed specifically to rationalise work. Grounded in formal execution semantics and workflow patterns, BPMN allows system architects to model task dependencies, concurrency (gateways), exception handling, and resource assignments without ambiguity. By rendering the invisible structure of operational work visible and measurable, BPMN enables systematic bottleneck analysis, throughput optimization, and formal verification — applying classical industrial engineering principles directly to modern digital and operational workflows.

5. Why is Formal Process Modeling Crucial for Mass-Series Sophisticated Production in the Digital Age?

The production of modern sophisticated products — whether physical automobiles, complex financial instruments, or global telecommunication services — requires coordinating thousands of interdependent activities across extended enterprise boundaries. In this environment, BPMN serves as the **blueprinting language** of the organization. Just as complex mechanical hardware cannot be manufactured at scale without CAD models and standardized engineering drawings (such as ISO or STEP standards),

enterprise-scale operations cannot achieve continuous, reliable execution without standardized process models acting as the single source of truth for both human actors and automated execution engines.

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6. What Role Does Model-Driven Engineering (MDE) Play in Reclaiming Process Industrialization?

To fully realize the rationalization promise of BPMN, process modeling must be integrated into a broader **Model-Driven Engineering (MDE)** framework. Under MDE principles, a BPMN diagram is not an idle post-hoc illustration; it is a primary engineering artifact that directly drives simulation, code generation, testing, and runtime orchestration. By linking BPMN XML models to underlying domain ontologies, data schemas, and system architectures (e.g., ArchiMate or SysML), organizations can automate the deployment of operational workflows while preserving strict traceability between high-level business goals and ground-level execution.

7. How Does BPMN Support Continuous Process Improvement (Lean & Six Sigma) at Enterprise Scale?

Process rationalization is not a one-time exercise but an iterative cycle. BPMN directly supports frameworks like **Lean Manufacturing** and **Six Sigma**

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feed runtime event data back into process models via **Process Mining** algorithms. This creates a closed-loop engineering system where real-world operational deviations are mapped against the reference BPMN model, allowing continuous refactoring and performance tuning.

Conclusion & Outlook

Questioning the relevance of BPMN process modeling often stems from losing sight of its original purpose: the **industrial rationalization of complex work.** Just as early twentieth-century industrial engineers unlocked unprecedented productivity by standardizing factory workflows, modern organizations require formal process blueprints to master the complexity of digital and multi-organizational ecosystems. BPMN 2.0 remains the premier open standard to achieve this rationalization. However, realizing this potential at scale requires moving beyond static diagrams toward executable, interoperable, and semantically validated models — a challenge that requires robust model transformation standards and dedicated semantic mediation architectures.

What are your thoughts on process rationalization?

- Is your organization leveraging BPMN as an engineering blueprint or treating it as static documentation?
- How do you bridge the gap between flexible knowledge work and the requirements of industrialized, repeatable execution?
- Which aspects of process industrialization and model-driven engineering would you like us to explore in upcoming articles?

Share your perspectives, questions, and industrial experiences in the comments below!

About the Author

I'm Dr. Nicolas Figay. My work as a researcher and practitioner centers on Continuous Operational Interoperability — finding robust ways to make heterogeneous systems, platforms, and extended enterprises work together seamlessly using open standards (such as BPMN, STEP/PLM, and UML). Throughout my career, my main goal has been to bridge the gap between formal specifications and real-world industrial execution, ensuring that data sovereignty and semantic integrity are preserved across organizational boundaries. I developed the open-source BPMN Semantic Mediator project, in connection with ArchiCG, as a practical effort to tackle these exact interoperability challenges, and I look forward to sharing these insights with you through this series.

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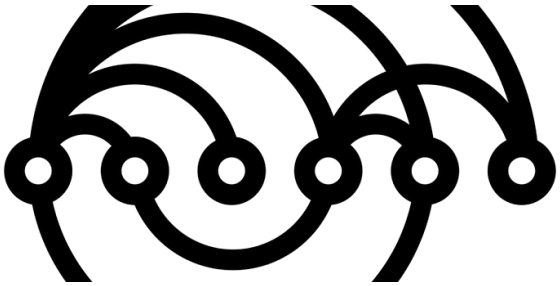
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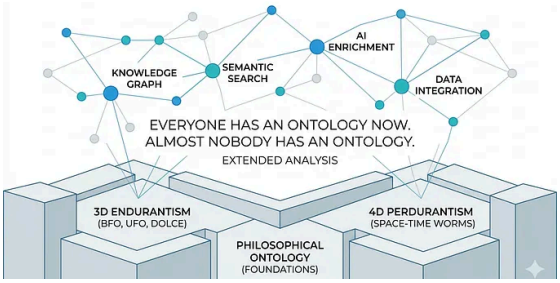


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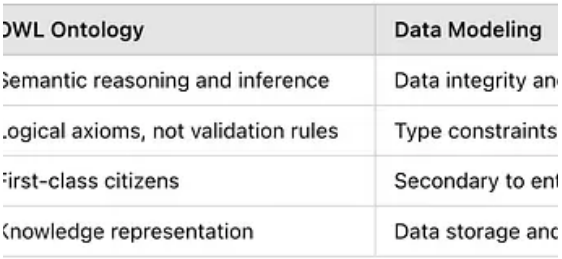


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Semantic reasoning and inference	Data integrity an
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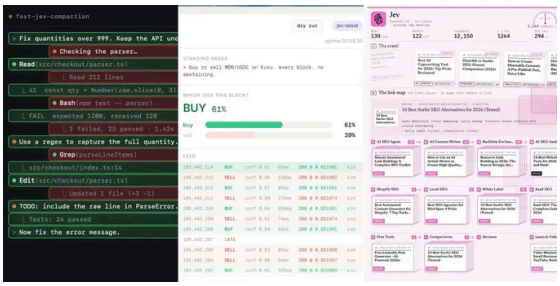
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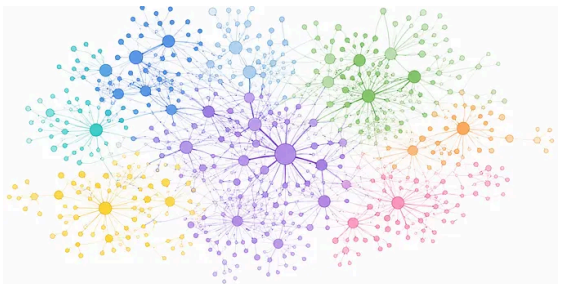


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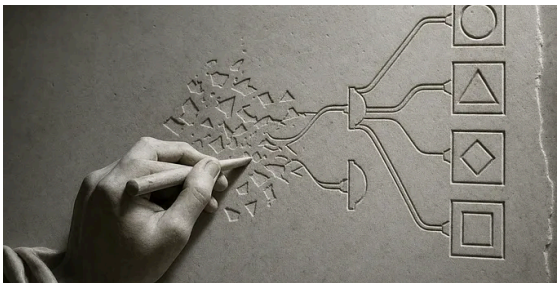


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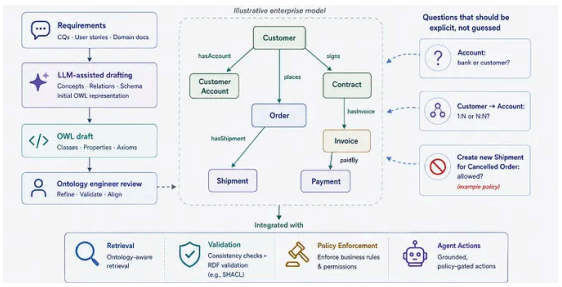
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